Zilin Chen

PhD Student, Mechanical Engineering | The University of Hong Kong | ArcLab
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EDUCATION

The University of Hong Kong | Nov 2026 - present

Full-time PhD in Mechanical Engineering | Expected completion: Oct 2030

Field of study: Autonomous UAVs | Supervisor: Prof. Peng Lu

School of Mechanical Engineering , Tsinghua University, Beijing, China Aug 2022 - Jun 2026

B.E. in Mechanical Engineering

GPA through sophomore year: 3.62/4.0 (Top 30%)

Standard English Tests: IELTS: Total 7.5 (Reading 8.5, Listening 8.5, Speaking 6.0, Writing 6.5)

Core Courses:

Fundamentals of Computer Program Design (A)

Mechanical Engineering Mechanics (1) (A+)

Physics for Scientists and Engineers B (2) (A)

Lab. of Physics B (1) (A-)

Selected awards:

Literary and Art Scholarship (2023, 2024)

Best social practice team Leader (2024)

Research interest: Automated control and trajectory planning, dynamics control

Department of Mechanical & Industrial Engineering , University of Toronto, Ontario, Canada Sep 2024 - Dec 2024

Exchange student in Mechanical Engineering

Major courses GPA for this semester: 3.9/4.0

Courses:

Kinematics and Dynamics of Machines (A-)

Fluid Mechanics I (A)

Circuits with Applications to Mechanical Engineering Systems (A+)

PROFESSIONAL EXPERIENCE

Beijing Keyi Technology Co., Ltd. | Former employee

Decision & Planning Algorithm Engineer | Joined Jul 2026 | Beijing, China

RESEARCH EXPERIENCES

Collaborative Robots Design | Research Assistant Feb 2024 - Dec 2024

Advisor: Ze Wang, professor and assistant dean at School of Mechanical Engineering, Tsinghua University

• Summarized current research progress on Motion Trajectory Planning of Two-arm Collaborative Robot

• Improved the stability on the end effector of a Two-arm Collaborative Robot

• Analyzed the workspace and constraints of the robot to prevent collisions between the two-arm Collaborative Robot

• Collaborated with a team of four to design different robot configurations to improve stiffness

Robot trajectory planning | Research Assistant May 2024 - July 2024

• Published on 9th IEEE International Conference on Advanced Robotics and Mechatronics , won the Best paper award in advanced robotics

• Planned the trajectory for a seven degree of freedom robot, based on Franka Robot

• Utilize charts to visually represent the changes in velocity, acceleration and jerk

• Employed multiple solutions to solve the time require ment to pass through the same set of waypoints, and compar e them through velocity, acceleration, and jerk

Intelligent Vehicle Design | Group Leader July 2024

• Led a team of three in designing an autonomous car-shaped vehicle capable of transporting objects from start to finish

• Integrated advanced features including line -following, obstacle navigation, destination location, and Bluetooth connectivity for remote control

• Acquired proficiency in utilizing PID control algorithms to fine-tune input parameters for precise motor control

• Utilized an STM32 microcontroller for vehicle control and Open MV for image capture (computer vision) , trajectory planning, and dynamic output management

UTAT Autonomous Drone Racing Team | Research Assistant Sep 2024 - present

Advisor: Hugh H.T. Liu, Professor & Director of the Centre for Aerial Robotics Research and Education, University of Toronto

• Helped with tests over ESC switching frequency and speed curve

• Built a simulator for a racing drone, mainly focus on the controller part , extract from the open-source project and convert it into a usable version for the onboard computer

• Planned the yaw-axis trajectory for the racing drone, under a given routine and gates , using CasADi as solver to solve for the optimal question

• Increasing the time for the drone to have a view of the gate, improving positioning accuracy

LEADERSHIP AND ACTIVITIES

Tsinghua University Symphony Orchestra | CCYL Tsinghua University Committee | Vice President Aug 2023 - July 2024

• Coordinated the orchestra to participate in the 7th National College Student Art Exhibition, and won the top prize

• Responsible for daily management of the orchestra and organizing the daily training

• Organized multiple concerts with at least 100 audience and 30 performers for each concert

Student Social Practice Team | CCYL Mechanical Engineering Committee | Project Leader Jan 2024 - Mar 2024

• Recruited 14 members on campus and invited 2 students from HUST and TJU to participate

• Visited multiple mechanical industry cooperations and communicated with frontline engineers

• Awarded the Golden Award for Social Practice in the Department, and personally awarded the Best Team Leader

SKILLS

Metal craftsmanship and industrial robot experiment skills

Programming Languages: proficient in C & Python & MATLAB, familiar with Java & C++

Office Applications: Microsoft Office, Photoshop, Auto CAD, SolidWorks, Multisim